

SCIBENCH ANONYMIZED REPLICATION DOSSIER

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Anonymous replication dossier: finite-temperature single-spin curves

Scientific objective

For one spin in a constant field along z , reproduce five families of thermal expectation-value curves. Compare the exact discrete quantum calculation, continuous coherent-state approximations, and classical limits. This task evaluates only the finite deterministic curves; stochastic spin-dynamics estimates and graphics are outside its scope.

Definitions

Let s be the spin quantum number, $\beta = 1/(k_B T)$, and use the Zeeman Hamiltonian $H = -g \mu_B B_z S_z / \hbar$. In the discrete basis, $S_z |s, m\rangle = \hbar m |s, m\rangle$, with $m = -s, \dots, s$. The exact reference expectation is

[EQUATION]

$$\begin{aligned} \langle S_z \rangle_q &= \\ & \frac{\sum_{m=-s}^s \hbar m \exp(\beta g \mu_B m B_z)}{\sum_{m=-s}^s \exp(\beta g \mu_B m B_z)}. \end{aligned}$$

Introduce coherent states indexed by complex z :

[EQUATION]

$$\begin{aligned} |z\rangle &= (1+|z|^2)^{-s} \\ & \sum_{p=0}^{2s} \binom{2s}{p}^{1/2} z^p |p\rangle, \\ d\mu(z) &= (2s+1)/\pi * dz/(1+|z|^2)^2. \end{aligned}$$

Map z to the unit-vector component

[EQUATION]

$$n_z = (1-|z|^2)/(1+|z|^2).$$

The leading moments needed by the expansion are

[EQUATION]

$$\begin{aligned} \langle z | S_z | z \rangle &= \hbar s n_z, \\ \langle z | S_z^2 | z \rangle &= (\hbar s n_z)^2 \\ &+ 2 \hbar^2 s |z|^2 / (1+|z|^2)^2. \end{aligned}$$

Higher powers contain the classical term $(\hbar s n_z)^k$ plus corrections of higher order in $1/s$. Generate the coherent-state approximation by expanding the operator exponential in β and retaining the requested number of correction orders for each logical curve.

Classical and coherent-state expectations

The classical-limit expectation can be evaluated as

[EQUATION]

$$\langle n_z \rangle_{cl} = \frac{\int_{-1}^1 n_z \exp(\beta g \mu_B s B_z n_z) dn_z}{\int_{-1}^1 \exp(\beta g \mu_B s B_z n_z) dn_z}.$$

For a coherent-state effective Hamiltonian H_{eff} , use

[EQUATION]

$$\langle S_z \rangle_{eff} = \frac{\int d\mu(z) \hbar s n_z \exp(-\beta H_{eff})}{\int d\mu(z) \exp(-\beta H_{eff})}.$$

A coarse low-temperature correction is

[EQUATION]

$$H_{eff_low} = -g \mu_B B_z s n_z + (1/2) g \mu_B B_z \sqrt{2s} \sqrt{1-n_z^2}.$$

For the high-temperature construction, start from the exact coherent-state integrand

[EQUATION]

$$F(\beta, z) = \exp(-\beta g \mu_B s B_z) ((\exp(\beta g \mu_B B_z) + |z|^2)/(1+|z|^2))^{(2s)}.$$

Expanding $\log(F)$ through third order in β gives the effective Hamiltonian

[EQUATION]

$$H_{eff_high} = -g \mu_B s B_z n_z - (1/4) \beta (g \mu_B)^2 s B_z^2 (1-n_z^2) + (1/12) \beta^2 (g \mu_B)^3 s B_z^3 n_z (1-n_z^2).$$

When the approximate coherent-state observable is used, apply the normalization required by its $\beta \rightarrow \infty$ limit, which is $s^2/(s+1)$, so the saturated observable has the correct scale.

Curve groups

Use the following fixed curve ordering:

- Group 1 ('curve_01'--'curve_05', $s=0.5$): classical limit; coherent-state series with one, two, and three retained correction orders; exact quantum. Use 500 uniformly spaced temperatures from 0.02 K to 5 K.
- Group 2 ('curve_06'--'curve_14'): classical curves for $s=\{0.5, 2, 5\}$; low-temperature curves for $s=\{0.5, 2, 5\}$; exact quantum curves for $s=\{0.5, 2, 5\}$. Use 100 uniformly spaced temperatures from 0.02 K to 5 K.
- Group 3 ('curve_15'--'curve_18', $s=2$): classical, first-order high-temperature, second-order high-temperature, and exact quantum curves. Use 100 uniformly spaced temperatures from 0.07 K to 5 K.
- Group 4 ('curve_19'--'curve_21', $s=2$): normalized coherent-state, unnormalized coherent-state, and exact quantum curves.

Use 50 uniformly spaced temperatures from 0.07 K to 5 K.

- Group 5 ('curve_22'--'curve_23', s=2): classical and exact quantum curves. Use 500 uniformly spaced temperatures from 0.02 K to 5 K. The divergent high-order curve with non-finite low-temperature values is intentionally excluded.

[MASKED: all reference curves and numerical comparisons are withheld.]